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Bereken de volgende limiet met de regel van de l'Hôpital $\lim_{x \to \frac{\pi}{2}} \dfrac{\tan \frac{x}{2} - 1}{\cos x}$

$$\overset{H}{=} \lim_{x \to \frac{\pi}{2}} \frac{\dfrac{1}{\cos^2 \frac{x}{2}} \cdot \dfrac{1}{2}}{-\sin x}$$

$$= \lim_{x \to \frac{\pi}{2}} \frac{1}{-\sin x \cdot \cos^2 \frac{x}{2} \cdot 2}$$

$$= \lim_{x \to \frac{\pi}{2}} \frac{1}{-\sin \frac{\pi}{2} \cdot \cos^2 \frac{\pi}{4} \cdot 2}$$

$$= \lim_{x \to \frac{\pi}{2}} -\frac{1}{\left(\dfrac{\sqrt{2}}{2}\right)^2 \cdot 2}$$

$$= \lim_{x \to \frac{\pi}{2}} -\frac{1}{\dfrac{2}{4} \cdot 2}$$

$$= -1$$

References